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Review of the Behavior of Geopolymer Cement Exposed to Acidic Environment Under CO_2 Sequestration Conditions

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Abstract

Geopolymer cement is a promising alternative to wellbore cement for applications involving acidic environments, particularly in CO_2 sequestration wells. Its unique structure characterized by low Ca/Si ratio, low calcium content, and a highly cross-linked aluminosilicate gel matrix, provides superior resistance to chemical degradation, reduced permeability, and stable mechanical properties under high temperature and pressure conditions. The paper provides a review of the recent research on geopolymer cement use for applications of CO_2 sequestration, it compares the chemistry of the wellbore cement (i.e. such as Class G cement) with that of geopolymer, it deeply compares that chemistry and investigate its impact on the stability of the cementitious materials developed in both systems in CO_2-rich environment. It also discussed microstructural behavior of both geopolymer based cement and wellbore cement in CO_2-rich environment and compare the long-term stability of both systems in such harsh conditions. This review shows that Geopolymer cement matrix has a unique chemistry as characterized by low calcium content which make it more stable in harsh conditions compared to wellbore cement. In terms of durability, geopolymer cement also showed long-term stability in acidic conditions. On the other hand, these geopolymer materials need standardization and also there is no any reported field application of such geopolymer cement especially in acidic environments such as CO_2 storage sites.

Introduction

CO_2 capturing and sequestration is a vital technology in mitigating global greenhouse gas emissions by capturing CO_2 and storing it into deep geological formations (Lackner 2003; Luo et al. 2022; Spellman 2016). For long term success, wellbore cement sheath integrity is crucial as any failure in cementing materials could lead to leakage of stored CO_2 back to the atmosphere (Takase et al. 2010; Udebhulu et al. 2024). The exposure of wellbore cement to carbonic acid, formed by CO_2 dissolution in water, creates a highly acidic and reactive environment. Conventional wellbore cement systems, particularly those based on ordinary wellbore cement are vulnerable to degradation under such conditions (Gan et al. 2024; Mahmoud et al. 2018, Mahmoud et al. 2022; Mahmoud and Elkatatny 2020; Omosebi, Ahmed, and Shah 2017;

Wang et al. 2024). Therefore, alternative cementing materials with enhanced chemical durability are being investigated to ensure secure and sustainable CO_2 storage.

Cement is used to seal the annular space between the casing and drilled formation to prevent fluid migration along the wellbore, this rule is critical for long term zonal isolation and environmental safety (Boukhelifa et al. 2004; Hu et al. 2025). However, when injected CO_2 contacted the formation water it forms carbonic acid (H_2CO_3) that when migrates around or through cemented sections can aggressively attack cementitious materials and react with its components through carbonation reaction. Cement carbonation and prolonged acid exposure can lead to leaching of calcium ions and formation of calcium carbonate which is later dissolved or destabilized composing structural integrity. This led to increased permeability, loss of mechanical strength, and microstructural damage of cement layer. Therefore, the chemical resilience and durability of cement under acidic conditions must be thoroughly understood and optimized for CO_2 storage applications (Mahmoud and Elkatatny 2019, 2020; Neves et al. 2024).

Geopolymer cement, a type of alkali-activated binder formed from aluminosilicate industrial by-products such as fly ash, slag, and metakaolin has gained attention as potential replacement for wellbore cement in aggressive environment (Mahmoud et al., 2024a; Nodehi and Taghvaee 2022; Sachdeva et al. 2025; Unis Ahmed et al. 2022). It is characterized by a low calcium content and chemically stable 3-dimensional Si-O-Al network, which enhances resistance to acid and thermal degradation. Unlike Portland based cement (OPC), geopolymer binders do not primarily relay on calcium-based hydration products and thus demonstrate superior durability under chemically aggressive conditions (Grengg et al. 2020a; Ojha and Aggarwal 2023; Wan-En et al. 2023). Their characteristic microstructural features, such as lower porosity and increased stability of gel phases, demonstrate their potential suitability in CO_2 sequestration wells, in which chemical integrity preservation becomes essentially critical.

In acidic mediums, as in those resulting during sequestration of CO_2, geopolymer cement shows considerably different characteristics than those of standard wellbore cement. The primary mode of degradation involves alkali ion leaching and partial matrix dissolution of the aluminosilicate fraction. Nevertheless, due to an absence of calcium hydroxide, which especially in wellbore cement is most susceptible to acid attack, and limited occurrence of leachable fractions, rates of dissolution remain comparatively low (Adufu et al. 2025; Ariyadasa et al. 2024; Grengg et al. 2020b; Shah, et al., 2024; Shokry, et al., 2024; Mahmoud et al., 2024b). Studies using scanning electron microscopy (SEM), X-ray diffraction (XRD) and infrared spectroscopy (FTIR) have shown minimal crack formation and relatively stable pore under extended acid exposure (Khalifeh et al. 2017a, 2017b; Li and Guo 2024). Understanding these chemical interactions is critical for Tailoring geopolymer formulations for under-ground applications.

Despite favorable outcomes observed in bench-scale tests, considerable gaps in knowledge exist regarding geopolymer cements' responses in realistic CO2 sequestration applications. Longer-term stability ofmicrostructure, interactions of geopolymer cements, and SO2 and temperature cycling effects remain poorly characterized. Reproducibility and overall performance also lack consistency due to variability in raw material composition and in curing regimes. Filling these gaps would be valuable both in terms of standardizing geopolymer cement systems and in increasing industry confidence.

The objective of this review is to integrate current knowledge and develop an understanding of areas still needing exploration to allow more widespread adoption of geopolymer cements in storage schemes. A comprehensive review of the microstructural behavior of geopolymer cement in the acidic conditions relevant to CO_2 sequestration is provided in this paper. It begins with discussion on the chemical and physical mechanisms governing geopolymer degradation in acid. This is followed by a comparative analysis with wellbore cement systems and an overview of characterization techniques used to assist microstructural changes. The review then highlights the influence of formulation parameters, curing conditions, and additives on acid resistance. Finally, it addresses key challenges, knowledge gaps, and future directions for optimizing geopolymer cement for long term acid resistant wellbore applications. This serves to inform research design and field implementation strategies.

Fundamentals of geopolymer cement

Chemistry and synthesis

Geopolymer cement represents a class of inorganic polymers formed through the alkali activation of aluminosilicate materials. The synthesis process involves the reaction between a solid precursor—typically rich in silicon (Si) and aluminum (Al)—and a highly alkaline activating solution. In contrast to wellbore cement, which gains strength through hydration of clinker minerals, geopolymerization is a chemical polymerization process that transforms solid aluminosilicate powders into a hardened binder, Common precursors include fly ash, metakaolin, and ground granulated blast furnace slag, which are industrial by-products rich in reactive amorphous phases. The alkaline activator, often composed of sodium hydroxide (NaOH) or potassium hydroxide (KOH) combined with sodium silicate (Na_2SiO_3), dissolves the Si and Al from the precursor to initiate the polymerization process. Fig. 1 illustrates the geopolymerization mechanism: under a highly alkaline activator solution (e.g. NaOH and sodium silicate), silicate and aluminate species are leached from the solid precursor particles and then undergo polycondensation to form a three-dimensional aluminosilicate network.

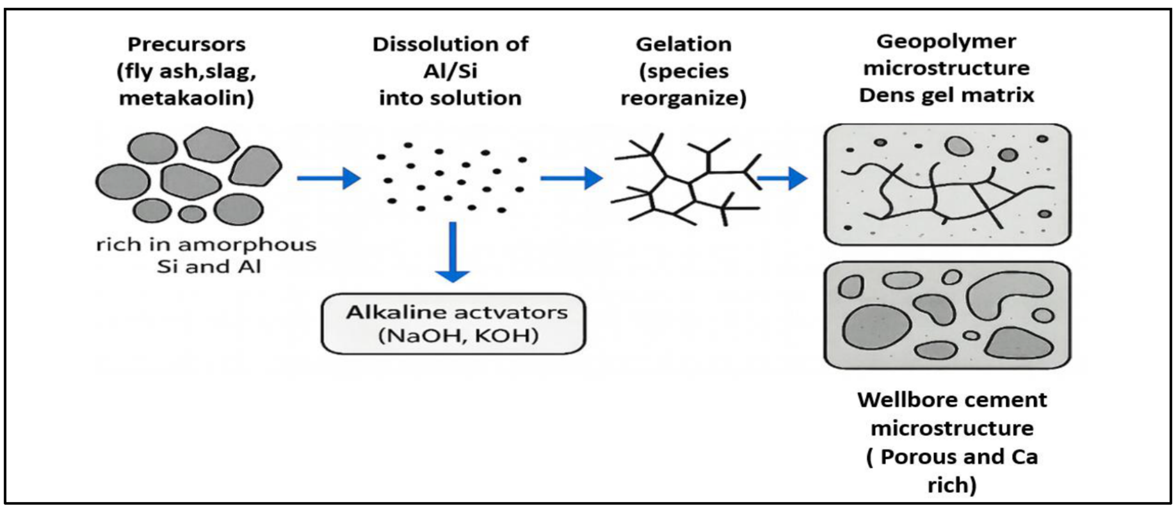

Figure 1—Schematic of Geopolymer Synthesis and Microstructure.

The chemical mechanism involves three stages: dissolution of reactive species, transport and orientation of dissolved ions, and polycondensation to form a rigid three-dimensional network of aluminosilicate gels. A simplified reaction can be represented by Eq. 1.

$$Al_2O_3 \cdot 2SiO_2 \cdot 2H_2O + NaOH \setminus Na_2SiO_3 \rightarrow Na^{+}(-Si-O-Al-O-)_n \cdot wH_2O \qquad \text{Eq. 1}$$

This process leads to the formation of sodium aluminosilicate hydrate (N-A-S-H) gel, which is the primary binding phase in most geopolymer systems. Unlike ordinary wellbore cement, which relies on calcium silicate hydrate (C-S-H) gels, geopolymer cement is generally calcium-free or low in calcium, making it more stable in acidic and high-temperature environments. The chemistry of geopolymers allows for rapid setting and high early strength, with considerable flexibility in adjusting properties through variations in activator concentration, Si/Al ratio, and curing temperature.

Replacement of wellbore cement by geopolymer is expected to significantly reduce the CO_2 emission, since geopolymer production does not depending on using material that need calcination or heating at very high temperature similar to the case of wellbore cement production. Therefore, geopolymer could be an attractive option talking about environmentally sensitive applications.

Typical Microstructure

The microstructure of geopolymer is characterized by a **dense, highly cross-linked gel** with micro- to nano-scale pores that are generally discontinuous, and it is fundamentally different from that of wellbore cement due to its distinct chemical composition and reaction mechanism. In wellbore cement systems, hydration leads to the formation of calcium silicate hydrate (C-S-H) gel, calcium hydroxide (CH), and ettringite. Although these hydration products are responsible for providing he cement matrix with the needed mechanical strength, but especially in aggressive environment they are also responsible about the cement matric chemical vulnerability, especially the hydration products with high Ca/Si ratio.

In contrast, the geopolymer and depending on the activator used can develop a highly cross-linked 3D framework of N-A-S-H or K-A-S-H, this formed gel is characterized by a very low Ca/Si ratio, and it usually forms a dense and coherent network of Si-O-Al bonds which is more chemically stable then wellbore cement hydration products and capable of resisting dissolution and cracking in acidic environment. Microstructural investigations using techniques such as scanning electron microscopy (SEM), X-ray diffraction (XRD), and mercury intrusion porosimetry (MIP) have showed that geopolymer matrices typically have lower porosity and finer pore size distribution in compare to wellbore cement.

For instance, SEM images often show a more compact and homogenous matrix in geopolymer pastes, with fewer microcracks and reduced connectivity between pores. XRD patterns demonstrate the dominance of amorphous or semi-crystalline gel phases, lacking the crystalline calcium-based phases typical of wellbore cement. These features contribute to a lower permeability and enhanced durability under acidic and high-temperature conditions, Table 1 compares the microstructural characteristics of geopolymer cement and wellbore cement.

Table 1—The microstructural characteristics of geopolymer cement and wellbore cement.

Property	Geopolymer Cement	Wellbore Cement (i.e. Class G Cement)
Main binding phase	N-A-S-H / K-A-S-H gel	C-S-H gel, $Ca(OH)_2$, ettringite
Porosity	Low	Moderate to high
Pore size distribution	Fine and well-distributed	Broad, interconnected
Microcracking tendency	Low	Moderate to high
Resistance to acid	High	Low

These microstructural attributes position geopolymer cement as a superior alternative for aggressive environments, including those encountered in CO_2 sequestration wells. SEM analysis further illustrates the typical microstructure of fly ash–metakaolin based geopolymers. At high magnification, the matrix reveals a combination of unreacted fly ash spheres, residual crystalline phases such as mullite needles, and a continuous geopolymer gel that binds the system together, as indicated in Fig. 2. At intermediate scale, the surface morphology shows a heterogeneous texture with numerous embedded spheres and visible voids, reflecting the partial dissolution of precursor particles and the development of a fine pore network. At lower magnification, the overall structure appears as a dense, consolidated framework where the gel phase integrates remaining particles and distributes porosity throughout the matrix (Barbhuiya et al. 2022). Together, these images provide a representative view of geopolymer microstructure: a complex assemblage of gel phases, unreacted precursors, and pores that underpins the durability and performance characteristics discussed earlier in this review.

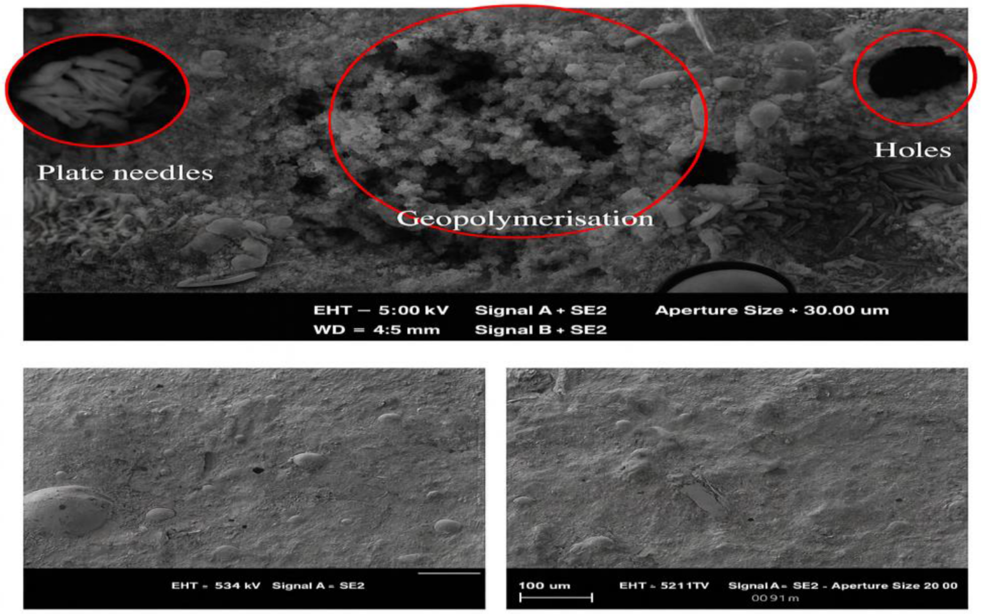

Figure 2—SEM Images for Different Geopolymer Systems (Barbhuiya et al. 2022).

Mechanical and Thermal Properties

Geopolymer cement demonstrates excellent mechanical properties, making it suitable for structural and downhole applications. Compressive strengths typically range between 30–80 MPa, depending on the type of precursor, activator concentration, and curing conditions. Geopolymers also exhibits high early strength often achieving over 70% of their ultimate strength within 24 hours under elevated curing temperatures their microstructural results in low permeability and essential feature for sealing well bores and preventing fluid migration.

Geopolymer binders are capable of withstanding high temperature often remaining structurally sound even above 400°C. This performance far surpasses that of wellbore cement, which typically begins to deteriorate between 200 and 250°C due to the thermal breakdown of CSH hydrates and the decomposition of CH. Aluminosilicate framework characteristic of geopolymer materials maintains it in form and function under these circumstances, and so these binders find extensive suitability for long-term exposure in applications such as geothermal wells and injection wells for CO_2.

As an added advantage to their potential for resistance to elevated temperatures, geopolymers also have good insulation properties and excellent fire resistance. Their expansion resistance in dimensions upon heat treatment, as well as their relatively small thermal expansion coefficient, minimizes the likelihood of cracking in temperature changes. Though their tensile and flexural strengths can be lower than in classical systems of concrete, these mechanical deficiencies can actually be somewhat increased through reinforcement in the form of reinforcing fibers or through nano-scale admixtures such as nano-silica and nano clay.

In regards to longevity, geopolymer cement presents superior resistance to negative chemical surroundings. It exhibits excellent resistance to acidic mediums, sulfate salts, and chloride invasion. This superior chemical resistance, coupled with thermal stability, renders geopolymer cement a sure bet in harsh downhole applications, particularly in cases such as in sequestration wells in CO_2, in which materials face persistent exposure to high heat, pressures, and reactive fluids.

Environmental and Sustainability Aspects

Geopolymer cement presents great environmental advantages over conventional wellbore cement, as it contains less embodied energy and less greenhouse gas emissions during manufacturing. Manufacturing of wellbore cement involves calcination of limestone at more than 1400°C and generates high volumes of CO_2 both from burning fuels and from calcium carbonate decarbonation. Comparatively, geopolymer binders involve industrial by-products, like fly ash, slag, and metakaolin, and do not involve treatment at elevated heat.

As such, the carbon footprint of geopolymer cement can be lowered as much as 80% in relation to traditional wellbore cement. This marked reduction is particularly valuable in carbon-sensitive applications, such as in CO_2 sequestration, in which material sustainability directly aligns with the initiative's overall environmental goals. Beyond this, geopolymer technologies can also enable waste valorization through the integration of materials otherwise sent to landfill, such as Class F fly ash or steel slag.

In addition to its potential for reduction of emissions, geopolymer cement can immobilize heavy metals and other toxic elements in its durable matrix. This property has been examined as an option for environmental remediation and for encapsulation of radioactive wastes, and it can also enhance the environmental effectiveness of wellbore seals in perforated geologically contaminated formations.

Life cycle assessment (LCA) research has been regular in showing geopolymer cement to be superior in terms of global warming potential, energy consumption, and consumption of raw materials. While challenges related to standardization, long-term field validation, and raw material variability remain, the environmental and performance benefits make geopolymer cement a highly attractive option for next-generation wellbore construction and CO_2 storage applications.

Acidic Environment in CO_2 Sequestration Wells

Origin of Acidic Conditions

Injection of supercritical CO_2 into depleted hydrocarbon reservoirs or deep saline aquifers initiates a series of complex geochemical reactions. Firstly, when CO_2 contacted the formation brine, it will dissolve and form carbonic acid which is a weak acid that will lead to reducing the pH of the formation brine. This process is governed by the following reactions:

As a result of this decrease in the pH of this formation brine the materials conducted by the acidic fluids will start to be affected including the spontaneous matrix behind the casing. This acidic environment could also be intensified because of the high pressure and temperature usually experienced in the CO_2 sequestration sites. Therefore, the long-term instability of materials in such environment is very critical to mitigating the scape of the stored CO_2. Usually traditional cement systems, such as OPC-based systems, are highly vulnerable to deterioration because of the carbonic acid that is usually present in CO_2 rich environment because of its high ability to lead to calcium leaching and loss of structural integrity of the cement. Understanding the origin and nature of the acidic environment is very critical for evaluating the long-term durability of alternative materials such as geopolymer which is recently proposed as a resilient solution for long term CO_2 containment.

Geochemical Conditions in Deep Formations

The geochemical environment in deep formations targeted for CO_2 sequestration is characterized by high pressures (typically >100 bar), elevated temperatures (ranging from 40°C to 150°C), and brine compositions that vary widely in salinity and ionic content. Under these conditions, the dissolution of CO_2 into brine significantly reduces pH levels, often reaching values below 4. The low pH environment creates a chemically aggressive medium that can interact with wellbore materials, including cement and casing steel.

The mineralogy of the host rock and formation water chemistry play a crucial role in buffering or exacerbating the acidity of the system. Carbonate-rich formations may temporarily buffer acidity through neutralization reactions; however, this also leads to the dissolution of carbonate minerals and potential porosity increase. In contrast, silicate-dominated formations provide limited buffering capacity and allow acidic fluids to persist longer. The geochemical evolution of these systems is further complicated by mineral dissolution-precipitation reactions, sorption processes, and redox conditions.

Geochemical modeling tools, such as PHREEQC and TOUGHREACT, are often used to simulate the behavior of CO_2-brine-rock interactions and predict long-term impacts on cement integrity. These models highlight the importance of selecting chemically stable cementitious materials capable of resisting acid attack. Geopolymer cements, with their low calcium content and highly polymerized aluminosilicate framework, offer enhanced resistance to acid-induced dissolution compared to wellbore cement. Evaluating their performance requires a clear understanding of the prevailing geochemical environment, which sets the stage for laboratory testing and field trials.

Impact of CO_2-Rich Environments on Cement Integrity

Exposure to CO_2-rich environments poses significant risks to the integrity of cementitious materials used for wellbore isolation. For wellbore cement systems, the most critical degradation mechanism is carbonation, where carbonic acid reacts with calcium hydroxide (CH) and calcium silicate hydrate (C-S-H) to form calcium carbonate ($CaCO_3$) and silica gel. These reactions are initially beneficial, as the formation of $CaCO_3$ can fill pores and reduce permeability. However, prolonged exposure leads to the leaching of calcium and the eventual breakdown of the cement matrix:

Carbonation of portlandite (calcium hydroxide) is shown in Eq. 2 and decalcification of calcium silicate hydrate (C–S–H) is explain by Eq. 3.

$$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O \qquad \text{Eq. 2}$$

$$C - S - H + CO_2 \rightarrow SiO_2 \cdot H_2O + CaCO_3 \qquad \text{Eq. 3}$$

These reactions result in increased porosity, loss of mechanical strength, and microcracking, thereby compromising zonal isolation and facilitating CO_2 leakage pathways. Additionally, the precipitation of secondary minerals can lead to volume changes and stress development within the cement sheath.

Geopolymer cements, on the other hand, exhibit significantly better performance in CO_2-rich environments due to their different chemistry and microstructure. The absence of reactive calcium phases eliminates carbonation as a major degradation route. Instead, the primary concern is the leaching of alkali ions and partial depolymerization of the aluminosilicate network. However, studies have shown that the rate of degradation is much slower and the structural changes less severe compared to wellbore cement.

Evaluating the long-term integrity of cement in CO_2 sequestration wells thus requires comprehensive knowledge of both chemical and mechanical degradation mechanisms. Laboratory tests, combined with microstructural analysis and performance modeling, are essential for predicting behavior under realistic field conditions and ensuring safe and effective CO_2 storage.

Microstructural Behavior of Geopolymer Cement Under Acidic Exposure

Chemical Degradation Mechanisms

Microstructure stability of geopolymer cement in acidic surroundings depends largely on the chemical composition as well as binding phase characteristics of geopolymer material. When compared to wellbore cement, which incorporates high calcium content and undergoes carbonation upon interaction with CO_2, geopolymer cement consists predominantly of alkali-activated aluminosilicate gels, i.e., N-A-S-H or K-A-S-H types. The properties of these gels include higher resistance to acidic corrosion due to inhibition of free calcium hydroxide and having a tightly cross-linked silicate network. However, erosion can still occur through leaching of alkali cations, for example, Na+ and K+, in combination with progressive depolymerization of aluminosilicate framework.

Under acidic conditions, protons (H+) in carbonic acid or other acid migrate into the pore solution in the cement and initiate ion exchange. H+ substitutes alkali cations in the geopolymer gel, which neutralizes the pore solution and initiates bond hydrolysis. These reactions can be summarized as in Eq. 4 and Eq. 5.

$$2Na^{+} + CO_2 + H_2O \rightarrow Na_2CO_3 + 2H^{+} \qquad \text{Eq. 4}$$

$$Si - O - Al \equiv +4H^{+} \rightarrow \equiv Si - OH + Al^{3+} + H_2O \qquad \text{Eq. 5}$$

This process releases soluble silica and aluminum that can diffuse out of the matrix, gradually weakening the load-bearing gel network. The rate and extent of degradation depend on pH, temperature, acid concentration, and the degree of gel polymerization. In most cases, acid exposure produces an alkali-depleted, partially depolymerized surface layer that advances slowly inward over time.

Fig. 3 illustrates this sequence. Acidic H+ first exchanges with alkali ions (Na+, K+) in the N–A–S–H gel, lowering pore solution alkalinity. This is followed by hydrolysis and dealumination, in which Si–O–Al linkages are disrupted and small amounts of dissolved silica and aluminum are released. As leaching proceeds, a silica-rich surface layer develops; it may exhibit slightly higher porosity and fine cracking, yet these changes are typically confined to near-surface regions. The figure emphasizes the stepwise, localized nature of the process: while acid exposure can roughen and weaken the outer layer, the bulk geopolymer matrix generally remains intact.

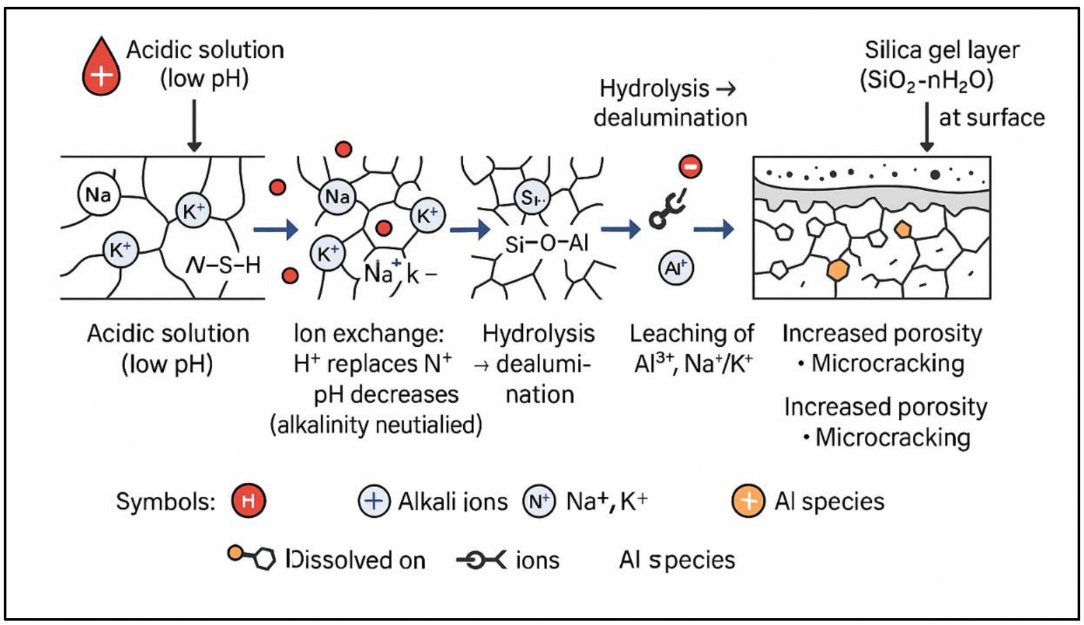

Figure 3—Mechanism of Geopolymer Cement Degradation in Acidic Formation.

Unlike wellbore cement, where chemical degradation rapidly compromises structural integrity, the effects in geopolymer cement are more localized and progressive. The presence of unreacted precursor particles and microstructural heterogeneity may also influence the degradation patterns. Despite this, the geopolymer matrix retains considerable integrity even after prolonged exposure, making it suitable for use in CO_2 sequestration wells and other acid-prone environments.

Microstructural Evolution and Analysis Techniques

Understanding how geopolymer microstructure evolves under acidic exposure is essential for assessing long-term performance. A suite of analytical techniques is typically employed to track changes in porosity, gel chemistry, and secondary products. Among these, scanning electron microscopy (SEM) offers direct, high-resolution evidence of surface alteration. In geopolymers, SEM commonly reveals only slight surface roughening, limited microcracking, and partial gel dissolution, while the interior matrix remains largely intact. By contrast, conventional wellbore cement often exhibits pronounced degradation, including deep cracking, broad decalcified rims, and porous, weakened layers that can extend millimeters into the section. Taken together, these observations indicate that geopolymer degradation is generally gradual and localized, whereas degradation in Portland-cement–based wellbore systems tend to be faster and more extensive.

X-ray diffraction (XRD) complements SEM by distinguishing between amorphous and crystalline phases. Because geopolymers are predominantly amorphous, XRD patterns typically retain the broad hump associated with the N–A–S–H gel, signaling preservation of the primary binding network. Minor crystalline products such as carbonates or zeolites may appear near exposed surfaces under acidic conditions, but the core gel signature remains largely unchanged. This persistence of the amorphous framework, even alongside limited surface crystallization, underscores the relative microstructural stability of geopolymers during acid exposure. Wellbore cement, on the other hand, shows dramatic changes: the portlandite peaks disappear and strong calcite peaks appear, signaling that calcium has been leached out and re-precipitated a clear marker of structural weakness. Fourier Transform Infrared Spectroscopy (FTIR) goes one level deeper by probing molecular bonds. In geopolymers, acid attack shifts peaks slightly and lowers their intensity, meaning some Si–O–Al bonds are broken. But these changes are relatively small compared with wellbore cement, where FTIR reveals major disruptions in C–S–H bonds and clear signs of network collapse.

Porosimetry techniques, including Mercury Intrusion Porosimetry (MIP) and nitrogen adsorption, measure changes in pore size and volume. After CO_2-rich acid exposure, geopolymers may show a slight increase in porosity and some finer pores forming, but these changes are usually minor. wellbore cement, in contrast, shows a big jump in porosity, coarser pores, and more interconnected pathways, all of which reduce its ability to act as a seal in the wellbore. These contrasting behaviors are summarized in Fig. 4, which compares the microstructural evolution of wellbore cement and geopolymer cements under acidic CO_2 exposure.

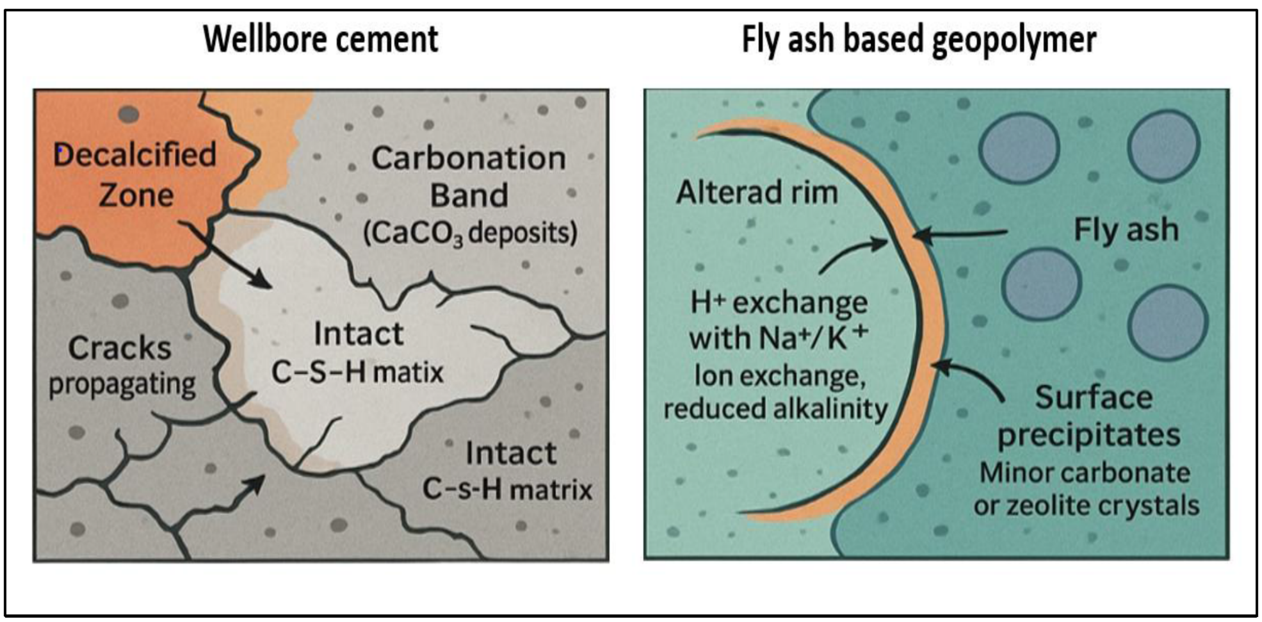

Figure 4—Comparative schematic of microstructural evolution in wellbore cement and geopolymer cements exposed to acidic CO_2 environments.

Overall, these analytical methods reveal that geopolymer cement maintains a relatively stable microstructure under acidic attack. While some deterioration occurs, especially at the surface, the internal matrix generally remains intact. This resilience is crucial for ensuring long-term sealing performance in CO_2 sequestration wells and supports the adoption of geopolymer systems in aggressive chemical environments.

Time-Dependent Performance and Durability

The long-term durability of geopolymer cement under acidic exposure is a critical parameter for its application in CO_2 sequestration wells. Short-term tests often show minor leaching and surface degradation in the first days or weeks of exposure. However, time-dependent studies consistently indicate that the rate of deterioration slows considerably over longer periods. This self-limiting behavior is attributed to the development of a silica-rich passive layer at the exposed surface, which reduces further ingress of aggressive species and helps protect the interior matrix.

geopolymers exposed to CO_2-saturated brine retained more than 80% of their compressive strength even after months, accelerated aging experiments at higher pressures and temperatures have reported similar results. This is in stark contrast to wellbore cement systems, where strength loss and permeability increase are significant. The denser matrix of geopolymer cement, combined with its lack of reactive calcium phases, plays a pivotal role in limiting the extent of acid penetration and internal damage.

Microstructural observations over time further reinforce the durability of geopolymers. In experiments under conditions such as 50 °C and 10 MPa CO_2 exposure, metakaolin-based geopolymers retained much of their unreacted granular precursors, suggesting that only a thin altered rim forms while the bulk structure remains intact (Zhang et al. 2025). FTIR and bond characterization studies report shifts in Si–O–Al or Si–O-T vibration bands in geopolymers, but these shifts are modest compared to the more severe bond disruptions reported in wellbore cement systems under similar stress (Yusuf. 2023). XRD results in multiple studies (including the circulating fluidized bed ash geopolymers) show retention of the broad amorphous gel hump, with only minor carbonate or zeolite peaks when exposed to CO_2 or aging environments, whereas wellbore cement commonly exhibits loss of portlandite in favor of strong calcite peaks under CO_2 attack (State-of-the-art review of geopolymer concrete carbonation, 2024).

Formulation and curing conditions strongly influence long-term performance. Geopolymers with higher Si/Al ratios tend to form more polymerized gels that are inherently more resistant to acid attack. Incorporating supplementary materials such as slag or nano-silica can refine pore structures, reduce permeability, and improve durability further. Elevated temperature curing promotes more homogeneous gel formation, producing matrices that maintain structural integrity even after prolonged exposure.

When contrasted with wellbore cement, the durability advantage of geopolymers becomes clear. Wellbore cement degradation is progressive: carbonation and decalcification deepen with time, porosity increases steadily, and permeability rises by up to two orders of magnitude after extended CO_2 exposure. Geopolymers, by comparison, show a plateau effect where degradation slows once the passive silica layer has formed, meaning strength and sealing capacity remain largely preserved even after months of acid exposure.

Despite these promising findings, knowledge gaps remain. Most durability studies cover timescales of weeks to a few months, whereas real sequestration wells must remain sealed for decades. Very few studies simulate more than one year of continuous exposure. Moreover, the combined effects of cyclic temperature and pressure loads, variable brine compositions (e.g., Mg^2+, H_2S-rich systems), and mechanical stresses on long-term geopolymer performance are not yet fully understood. Field-scale validation is essential to bridge the gap between laboratory data and in-well behavior.

Overall, the time-dependent behavior of geopolymer cement highlights its suitability for applications requiring prolonged resistance to acidic attack. While ongoing research is needed to fully characterize performance under field-like conditions, current evidence strongly supports the use of geopolymer systems for durable wellbore sealing in CO_2 sequestration and similar environments.

Geopolymer Cement vs. Portland Cement in CO_2-Rich Environments

In contrast to wellbore cement hydration, there is little or no calcium hydroxide involved in geopolymerization; geopolymer matrices are generally low in Ca content and are instead aluminosilicate gel-rich, imparting inherently different chemical behavior in the presence of CO_2 (Hajiabadi et al., 2023). One major advantage of geopolymers is their chemical stability in acidic and CO_2-rich conditions. Because of the low calcium phase content, there is minimal portlandite to be carbonated; indeed, geopolymer binders have shown exceptional resistance to acid attack and carbonation (they were originally noted to withstand aggressive acid environments in waste encapsulation applications (Nasvi et al., 2016). Uehara et al. (2010) directly compared wellbore cement and fly ash-based geopolymer samples in acidic brine and established that wellbore cement experienced extreme degradation (mass and strength loss), while the geopolymer experienced minimal changes in structure or compressive strength. Geopolymer cements also have significantly lower permeability and better long-term sealing ability under CO_2 exposure (Sathsarani et al., 2023). He report that even a plain fly ash geopolymer has gas permeability on the order of 10^(-16) m^2, which is an order of magnitude tighter than typical wellbore cement under similar conditions. By tailoring the geopolymer mix (e.g. adding slag), the permeability can be reduced even further; in one study, a fly ash geopolymer with 15% slag had a CO_2 permeability nearly 1000 times lower than that of a Class G cement sample. This drastic reduction in permeability is attributed to the dense gel microstructure and absence of large portlandite crystals, resulting in a finer pore network that is less connected (Nasvi et al., 2014). Mechanically, geopolymers can achieve high compressive strengths similar to oilwell cement, and tend to exhibit improved tensile behavior. For example, for the same 28-day compressive strength, the splitting tensile strength of a fly ash geopolymer may be greater than that of wellbore cement, because of the geopolymer's compact interfacial transition zones and dissimilar microcrack propagation behavior (Nasvi et al., 2016). This is useful since well cements for CO_2 service need to have mechanical integrity under pressure changes; the greater the tensile capacity, the lower the risk of fracturing. Geopolymers also experience almost no shrinkage on curing compared to wellbore cement's shrinkage – and so they bond well to steel casing and rock, without forming micro-annulus gaps (Nasvi et al., 2016). Improved cement–formation bond has been verified through field measurements: recent pilot implementations reported substantially better cement bond log response (amplitude reductions) using a geopolymer slurry compared to conventional cement, due to the lack of bulk shrinkage and cracking (Riyanto et al., 2023). Also, GPs are thermally more stable; they can withstand high downhole temperatures (and even short excursions to >300°C in thermal recovery operations) with little loss in strength, while wellbore cement may weaken as calcium phases convert or devolve at high temperature (this thermal stability is another motivation for investigating GPs for geothermal wells) (Kruszewski et al., 2021).

Conclusions

Geopolymer cement presents a promising alternative to Class G cement for applications involving acidic environments, particularly in CO_2 sequestration wells. Its unique structure characterized by low Ca/Si ratio, low calcium content and a highly cross-linked aluminosilicate gel matrix, provides superior resistance to chemical degradation, reduced permeability, and stable mechanical properties under high temperature and pressure conditions. Out of this review, the following points are concluded:

- The microstructural behavior of geopolymer cement under acidic exposure is notably favorable with minimal loss of strength and limited pore expansion compared to wellbore cement such as Class G cement.
- Time-dependent studies revealed that Geopolymer formulations maintain durability even after prolonged exposure to acidic carbonic acid and other aggressive fluids making them well suited or long term well bore sealing.

- Key factors influencing performance of geopolymer cement include the type and ratio of precursor materials, activator concentration, curing conditions, and the use of microstructural modifiers such as nanosilica and fibers.
- Many challenges remaining particularly in the area of field validation, material standardization, and compatibility with wellbore systems.
- Long-term field trials, coupled with advanced modeling and microstructural analysis, will be critical in confirming the stability of geopolymer cement for CO_2 sequestration and potentially expanding it is used across other demanding well bore applications.

Declaration

During the preparation of this work, the authors used ChatGPT in order to rephrase text for clarity, check grammar, and correct structural issues, and AI image generator was used to enhance the 2D images resolution. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication

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